(especially Chapters 6 through 8) will be familiar with much of this material, and should feel free to skim this chapter and move on to the next.

32 Remember, as noted in Chapter 9, even a puny magnet can overpower the pull of the entire earth's gravity and pick up a paper clip. Numerically, the gravitational force has about 10^{-42} times the strength of the electromagnetic force.

33 I might note that the proponents of another approach for merging general relativity and quantum mechanics, *loop quantum gravity,* to be briefly discussed in Chapter 16, take a viewpoint that is closer to the former conjecture—that spacetime has a discrete structure on the smallest of scales.

34 The relationship to mass arising from a Higgs ocean will be discussed later in the chapter.

35 Were you to count left, right, clockwise, and counterclockwise all separately, you'd conclude that the worm can move in four directions. But when we speak of "independent" directions, we always group those that lie along the same geometrical axis—like left and right, and also clockwise and counterclockwise.

36 Let me prepare you for one relevant development we will encounter in the next chapter. String theorists have known for decades that the equations they generally use to mathematically analyze string theory are approximate (the exact equations have proven difficult to identify and understand). However, most thought that the approximate equations were sufficiently accurate to determine the required number of extra dimensions. More recently (and to the shock of most physicists in the field), some string theorists showed that the approximate equations *missed* one dimension; it is now accepted that the theory needs *seven* extra dimensions. As we will see, this does not compromise the material discussed in this chapter, but shows that it fits within a larger, in fact more unified, framework.[20]

37 The more precise name for these sticky entities is *Dirichlet-p-branes,* or *D-p-branes* for short. We will stick with the shorter *p-brane.*